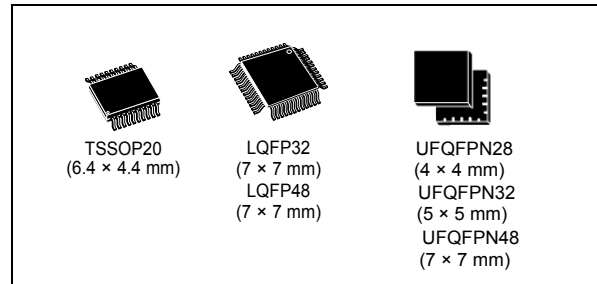


Arm® Cortex®-M0+ 32-bit MCU, 32 KB flash, 12 KB RAM,
2 x USART, timers, ADC, comm. I/Fs, 2-3.6 V

Datasheet - production data

Features

- Includes ST state-of-the-art patented technology
- Core: Arm® 32-bit Cortex®-M0+ CPU, frequency up to 48 MHz
- -40°C to 85°C/105°C/125°C operating temperature
- Memories
 - Up to 32 Kbytes of flash memory with protection
 - 12 Kbytes of SRAM with hardware parity check
- CRC calculation unit
- Reset and power management
 - Voltage range: 2.0 V to 3.6 V
 - Power-on / power-down reset (POR/PDR)
 - Programmable brownout reset (BOR)
 - Low-power modes: Sleep, Stop, Standby, Shutdown
- Clock management
 - 4 to 48 MHz crystal oscillator
 - 32 kHz crystal oscillator with calibration
 - Internal 48 MHz RC oscillator (±1 %)
 - Internal 32 kHz RC oscillator (±5 %)
- Up to 45 fast I/Os
 - All mappable on external interrupt vectors
 - All 5 V-tolerant
- 3-channel DMA controller with flexible mapping
- 12-bit, 0.4 µs ADC (up to 19 ext. channels)
 - Conversion range: 0 to 3.6 V
- 8 timers: 16-bit for advanced motor control, four 16-bit general-purpose, two watchdogs, SysTick timer
- Calendar RTC with alarm



- Communication interfaces
 - One I²C-bus interface supporting Fast-mode Plus (1 Mbit/s) with extra current sink; supporting SMBus/PMBus™ and wake-up from Stop mode
 - Two USARTs with master/slave synchronous SPI; one supporting ISO7816 interface, LIN, IrDA capability, auto baud rate detection and wake-up feature
 - One SPI (24 Mbit/s) with 4- to 16-bit programmable bitframe, multiplexed with I²S interface; two extra SPIs through USARTs
- Development support: serial wire debug (SWD)
- All packages ECOPACK 2 compliant

Table 1. Device summary

Reference	Part number
STM32C031x4	STM32C031C4, STM32C031F4, STM32C031G4, STM32C031K4
STM32C031x6	STM32C031C6, STM32C031F6, STM32C031G6, STM32C031K6